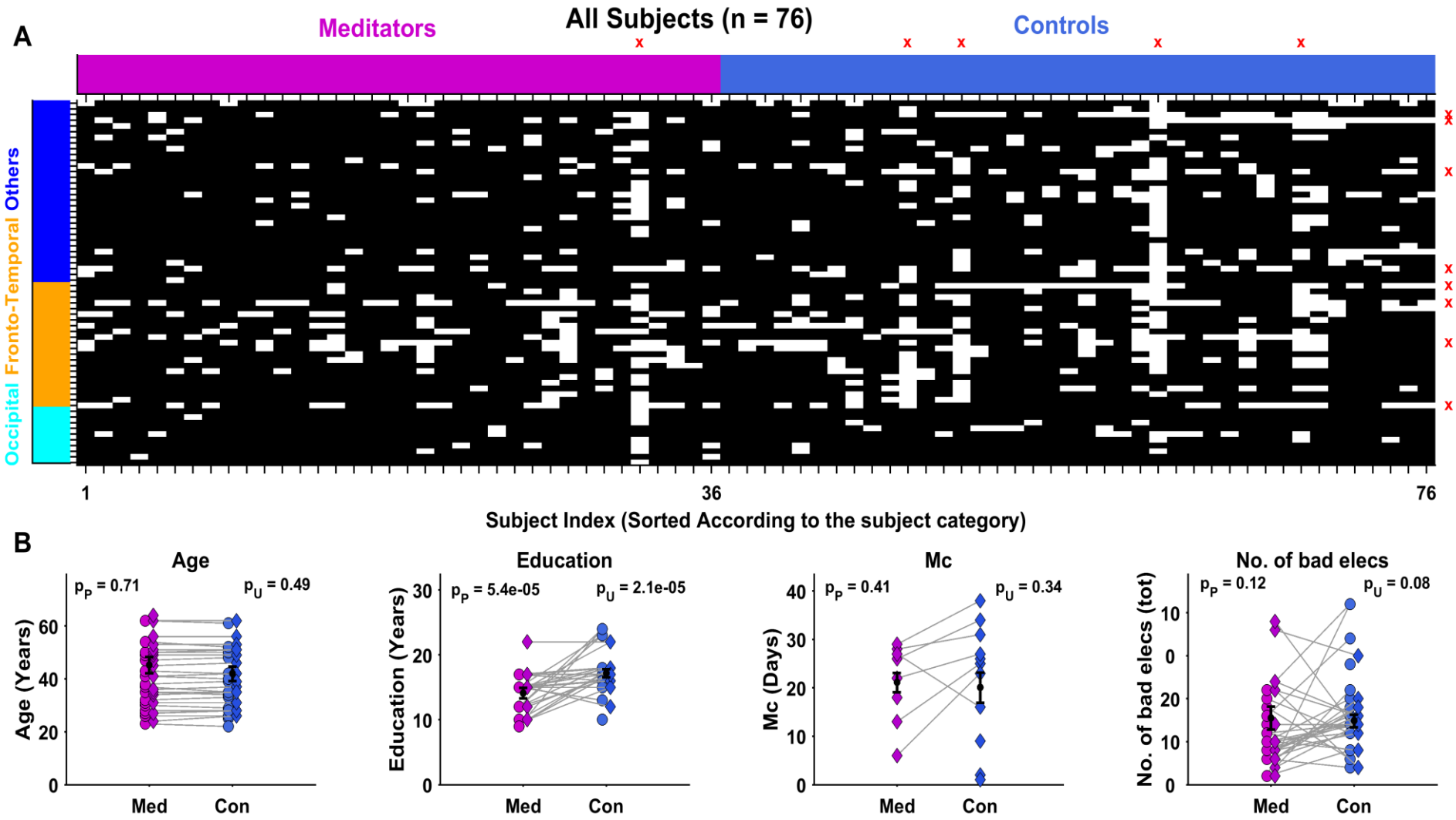
Study Details

- EEG data collected from 78 subjects
  - 38 advanced meditators (19 females) with at least 5 years of meditative practice
  - 40 control subjects (17 females) with no prior meditation experience
- Age spanning 20-65 years (uniform distribution)
- The meditators were adept Rajyoga practitioners of the Brahmakumaris (BK) tradition who practiced meditation with open eyes. A top-down coordination from headquarters of parent organization in Mt. Abu to meditation centers in Bengaluru city was done to ensure smooth participation of the subjects.
- Controls were recruited locally through word-of-mouth, broadcast-emails, and posters after matching their age (± 2 years) and gender with meditators. If we recorded from a control subject first, a suitable meditator was found in a similar way.
- For female subjects, we also tried to schedule the EEG recording such that the phase of the menstrual cycle was comparable to the meditator, since brain oscillations in the alpha/gamma band have been shown to depend on the menstrual phase.

Study Details

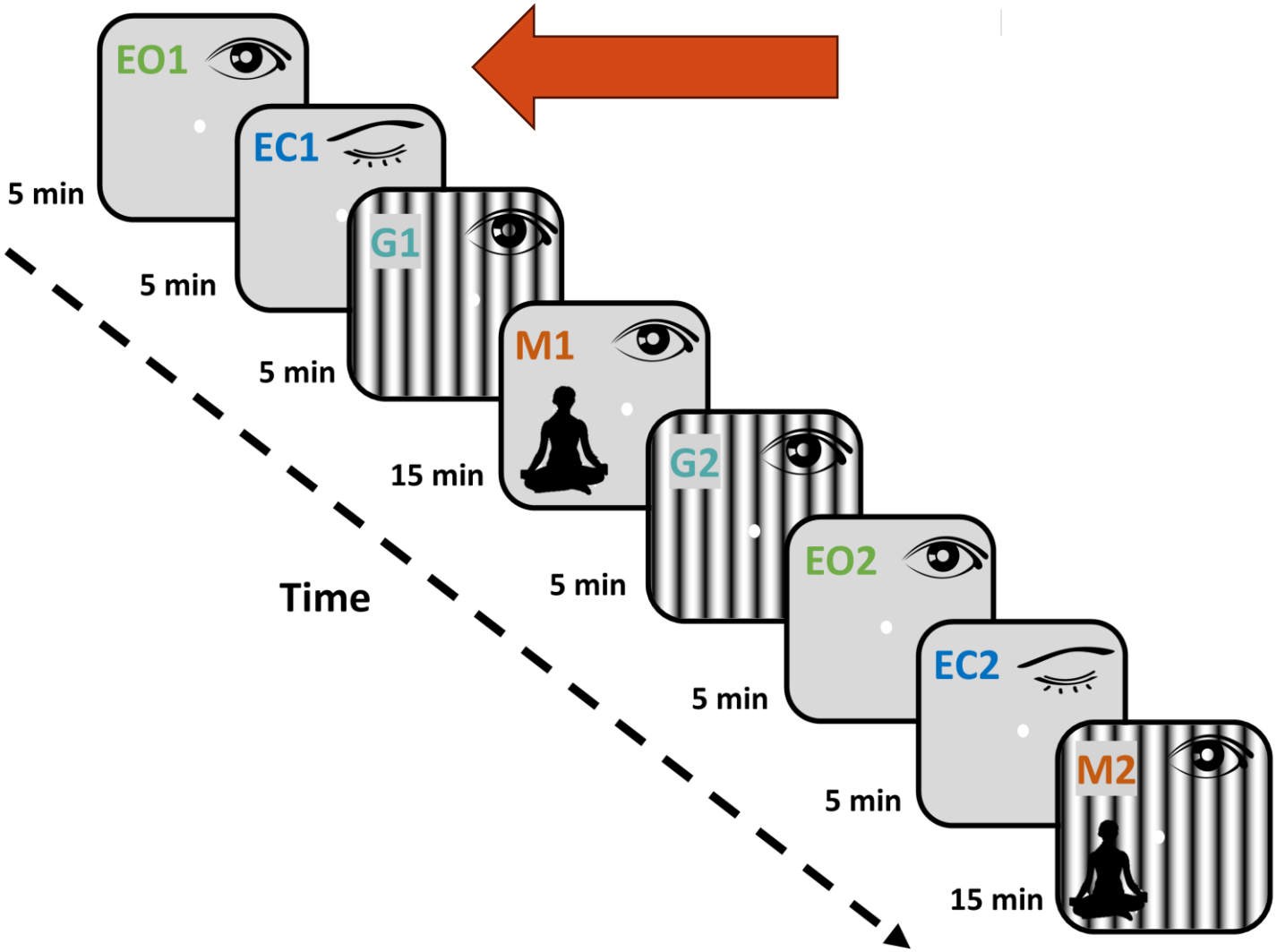
- For two meditators there was no control subject within ± 2 years, and hence these two subjects were removed.
- For the remaining 76 subjects, we first used a fully automated pipeline to find bad electrodes and bad stimulus repeats for each subject. We rejected 5 subjects (1 meditator and 4 controls) who had more than 24 (out of 64) bad electrodes.
- The usable data consisted of 71 subjects: 35 meditators (18 male and 17 female) and 36 controls (20 male and 16 female).
- Finally, we paired one control subject with each meditator. In case of multiple control subjects, we chose the one who had lower difference (in order of preference) in (i) age, (ii) education level, (iii) menstrual phase (for females) and (iv) EEG recording date.
- For 2 male and 3 female meditators, there was no control within 2 years of age difference. Therefore, out of 35 meditators, appropriate matching control was found for 30 pairs (16 males; 14 females).

Demographic Details



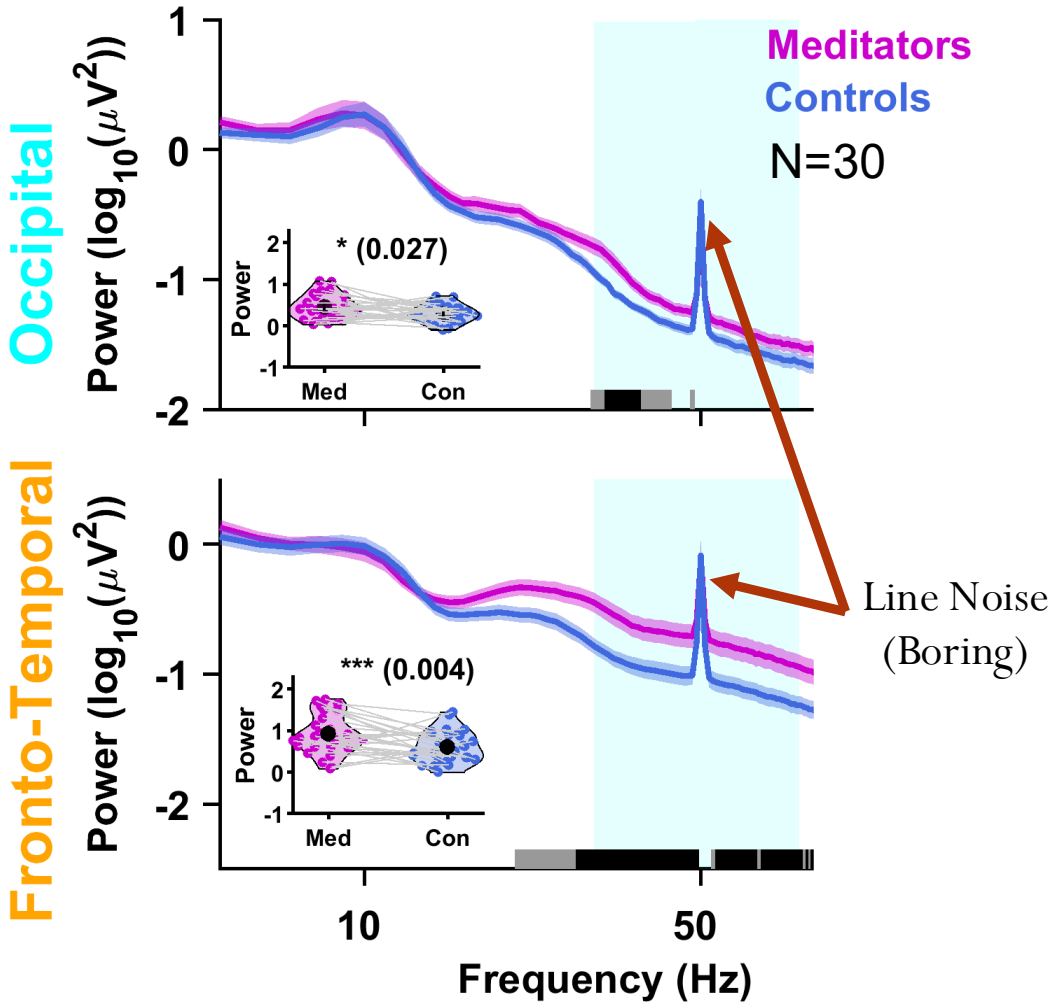
Biswas, Aggarwal, Sharma, Ray, 2024, bioRxiv

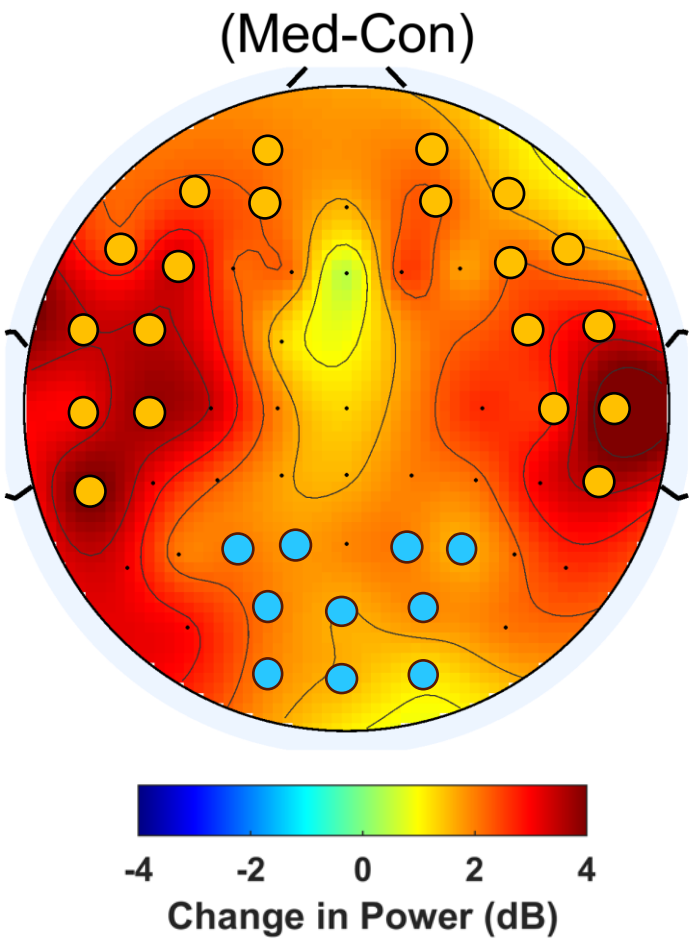
Non-Meditative States (EO1 and EC1)



Meditators have more spontaneous gamma even when they are not meditating

Eyes Open 1 (EO1)





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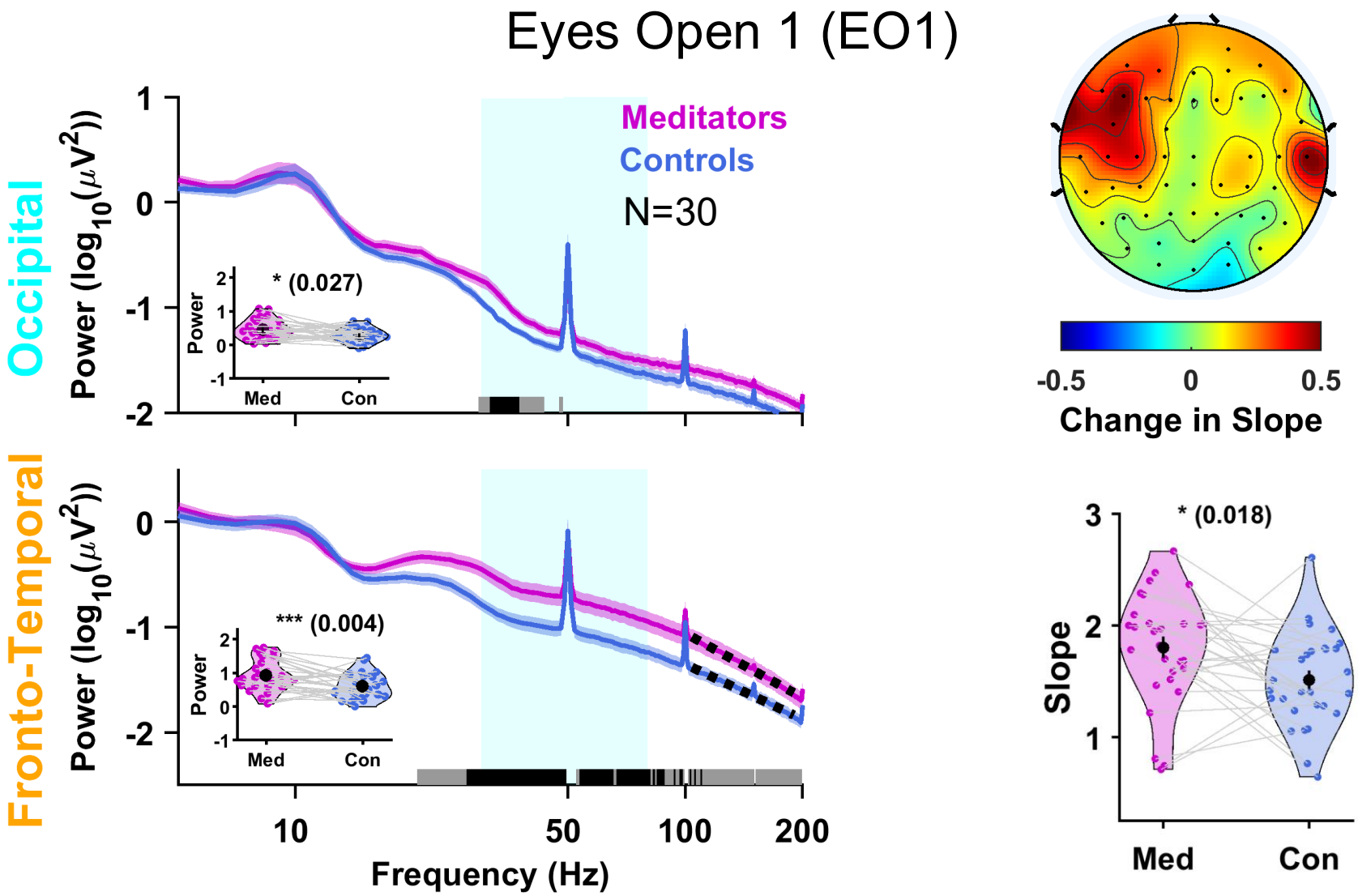
Meditators have steeper PSD slopes between 100-200 Hz in fronto-temporal areas



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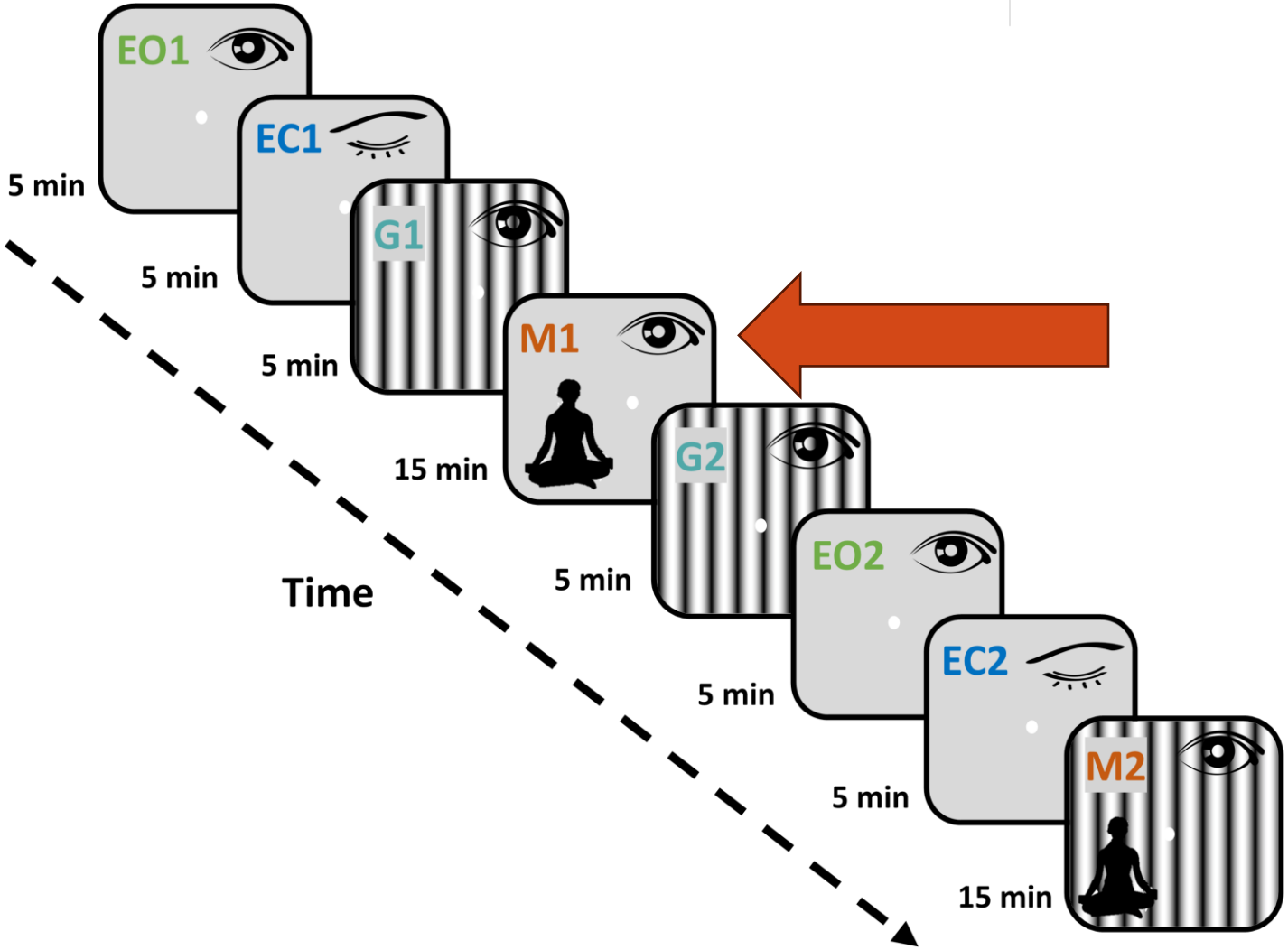
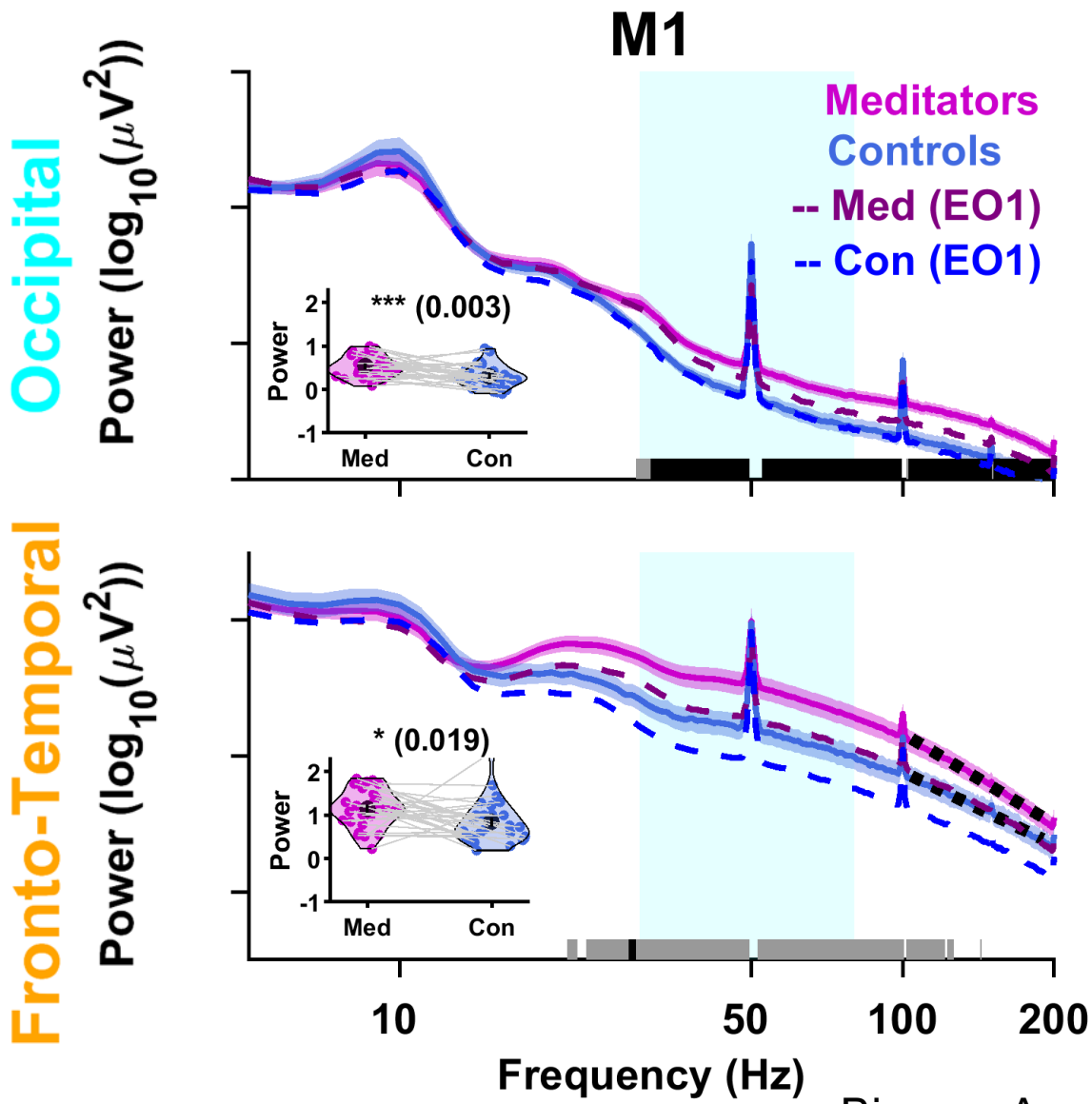
Stimulus-free Meditation (M1)



Broadband "Stimulus-free" Gamma increases for both meditators and controls



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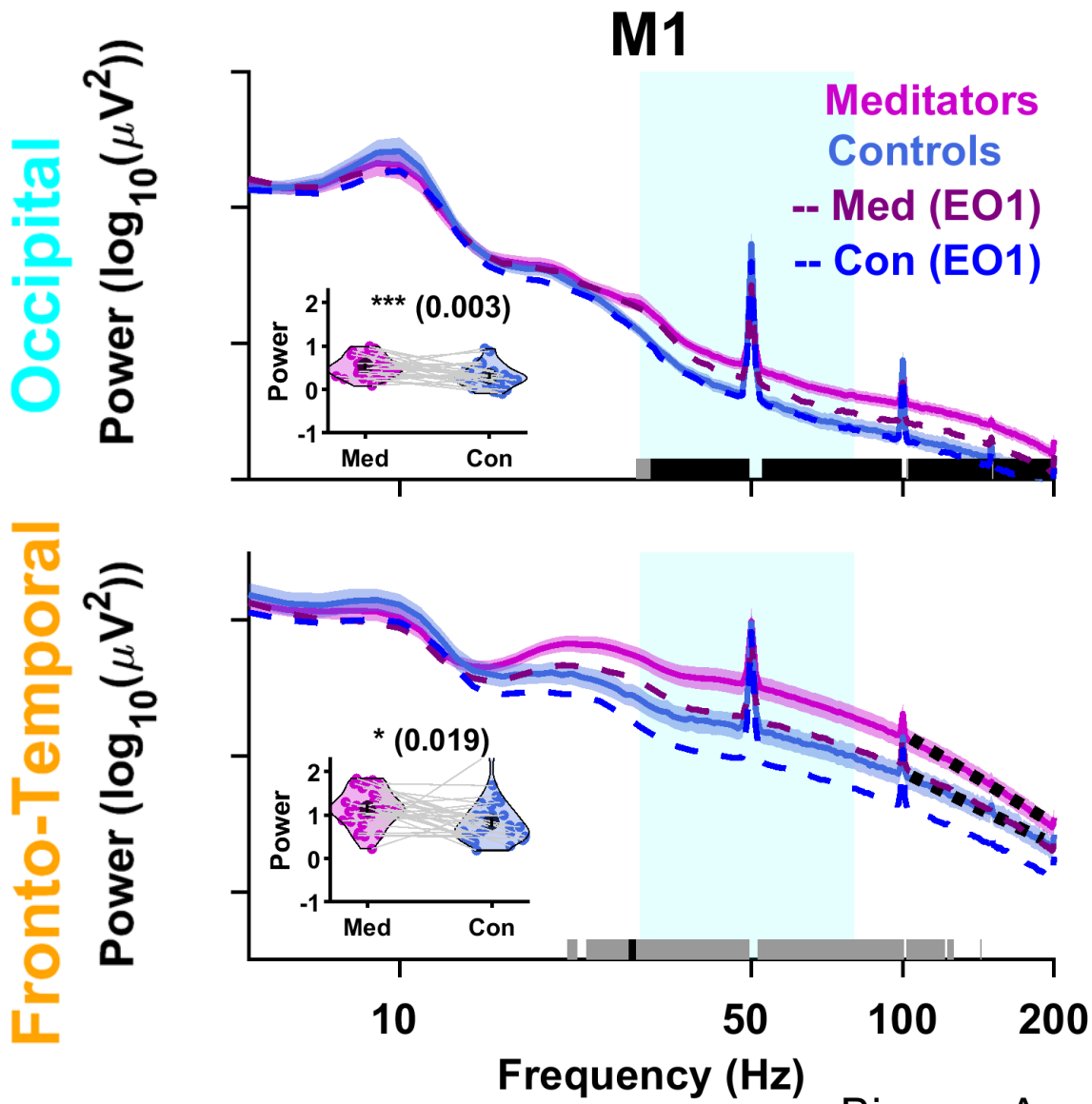
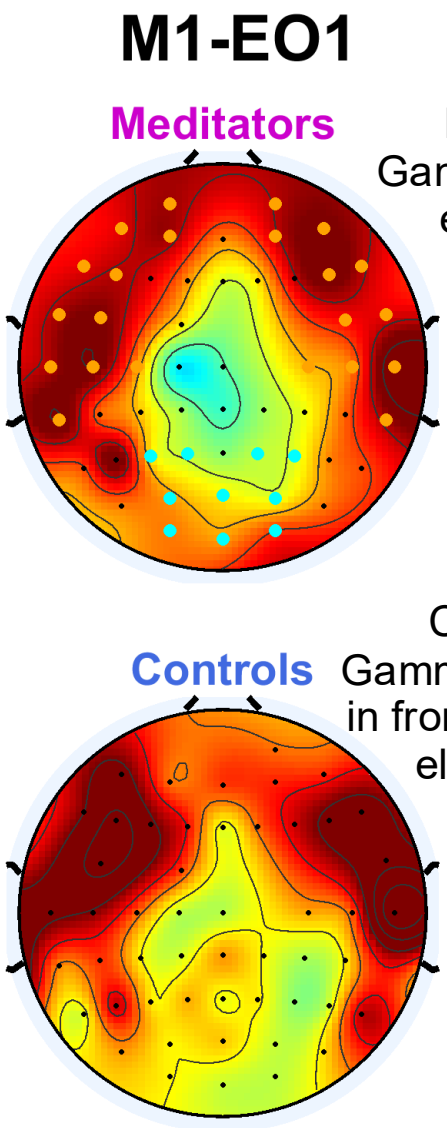
Broadband "Stimulus-free" Gamma increases for both meditators and controls





Meditators: Gamma increases everywhere

Controls: Gamma increases in fronto-temporal electrodes

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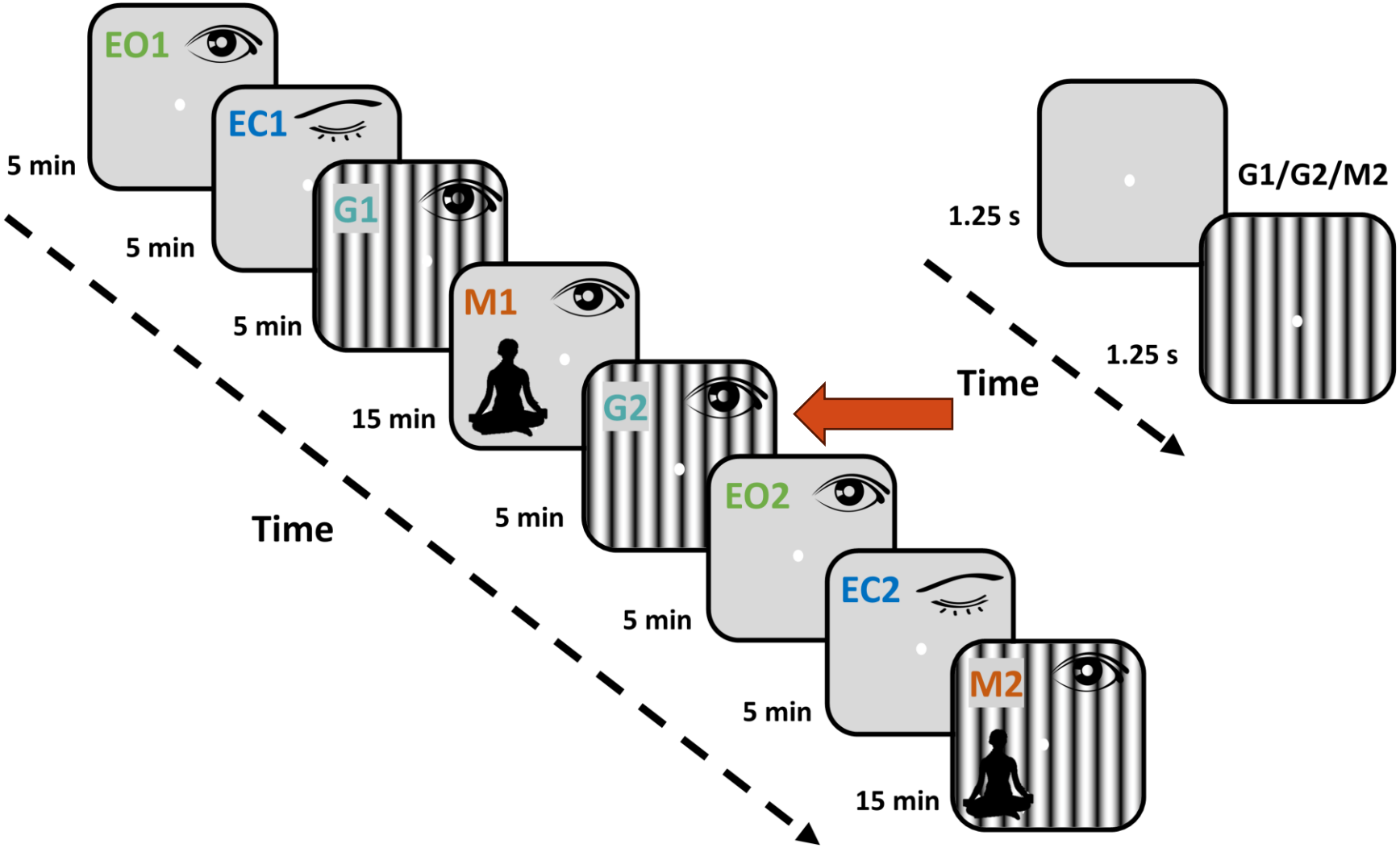
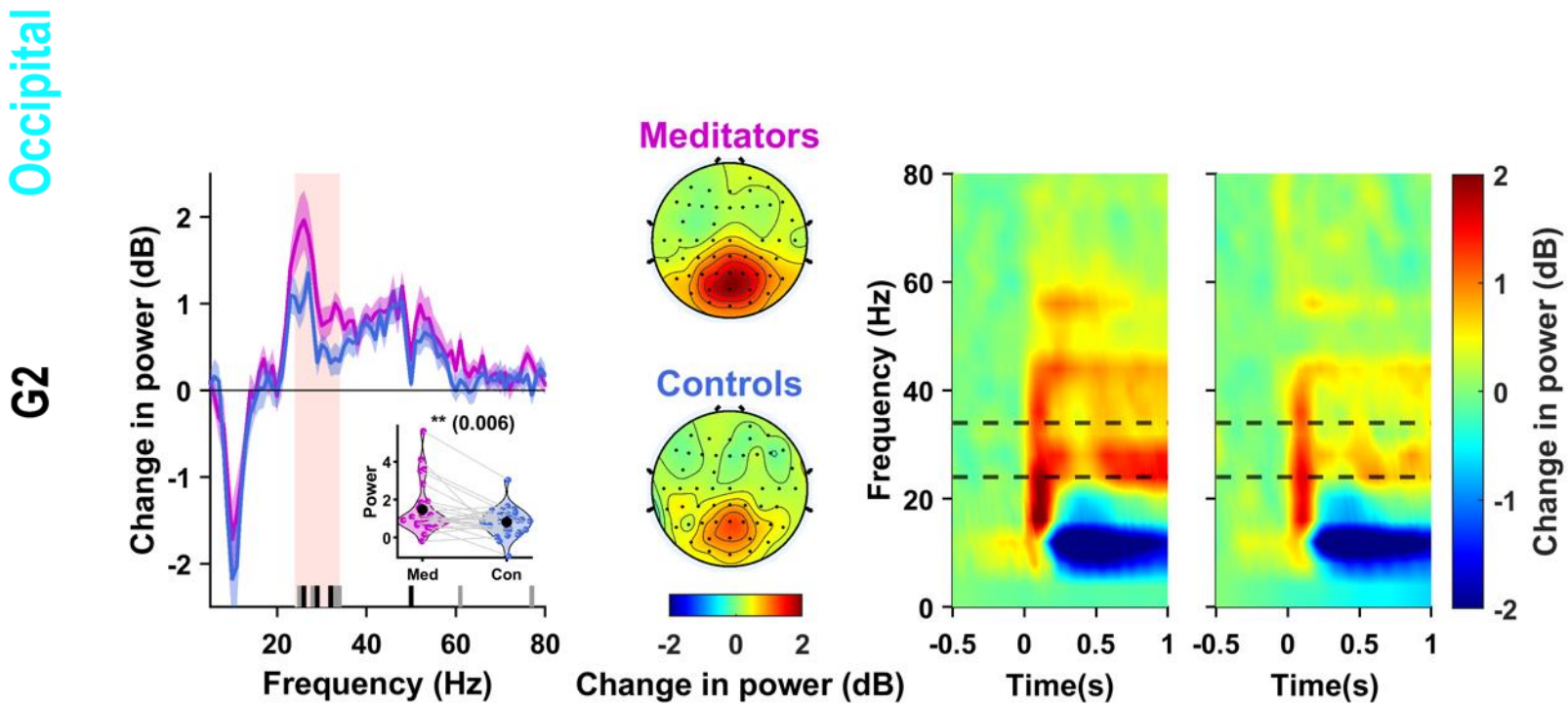
Stimulus-induced Gamma protocols



Slow-gamma, but not fast-gamma or alpha, is stronger in meditators



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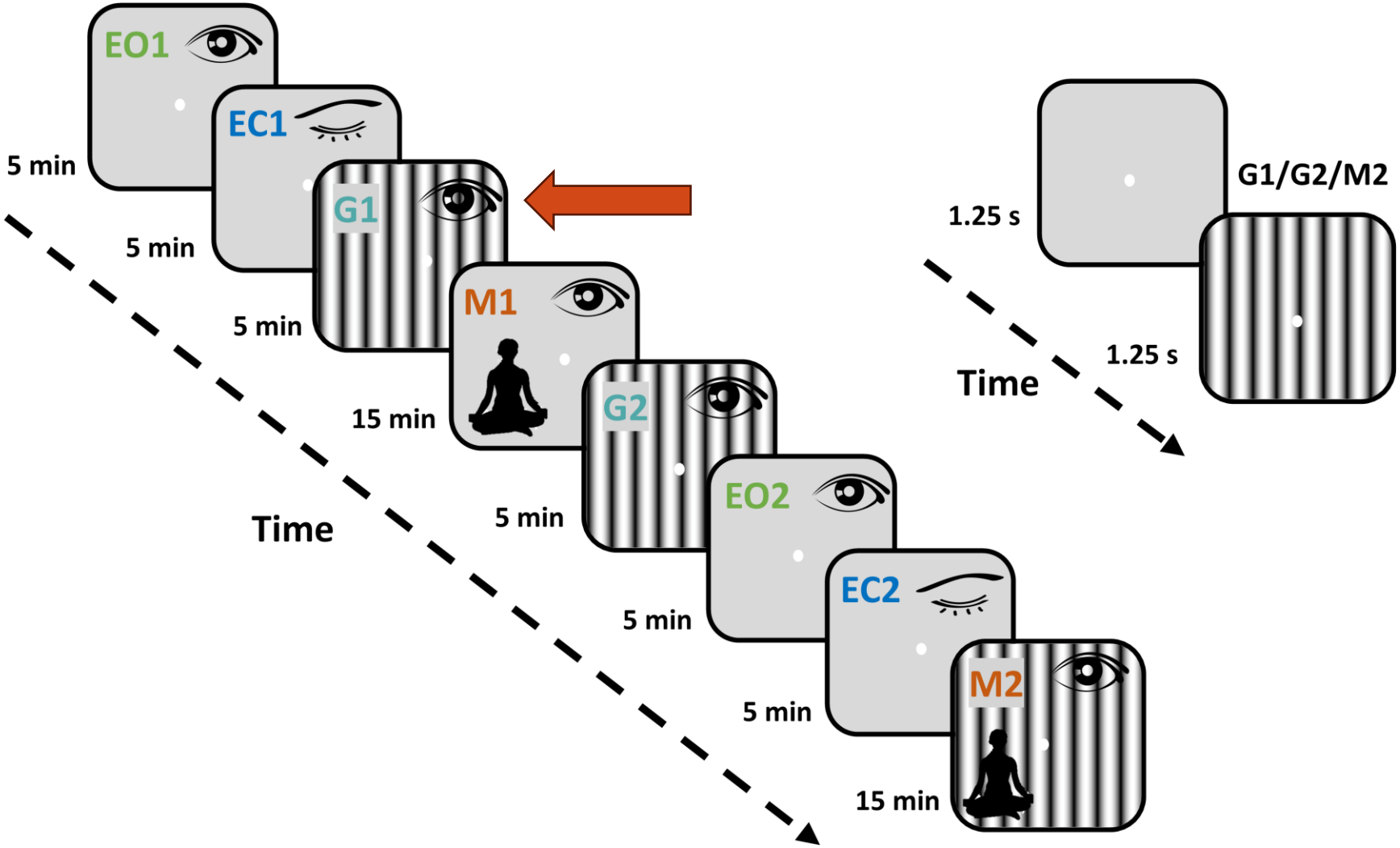
Stimulus-induced Gamma protocols



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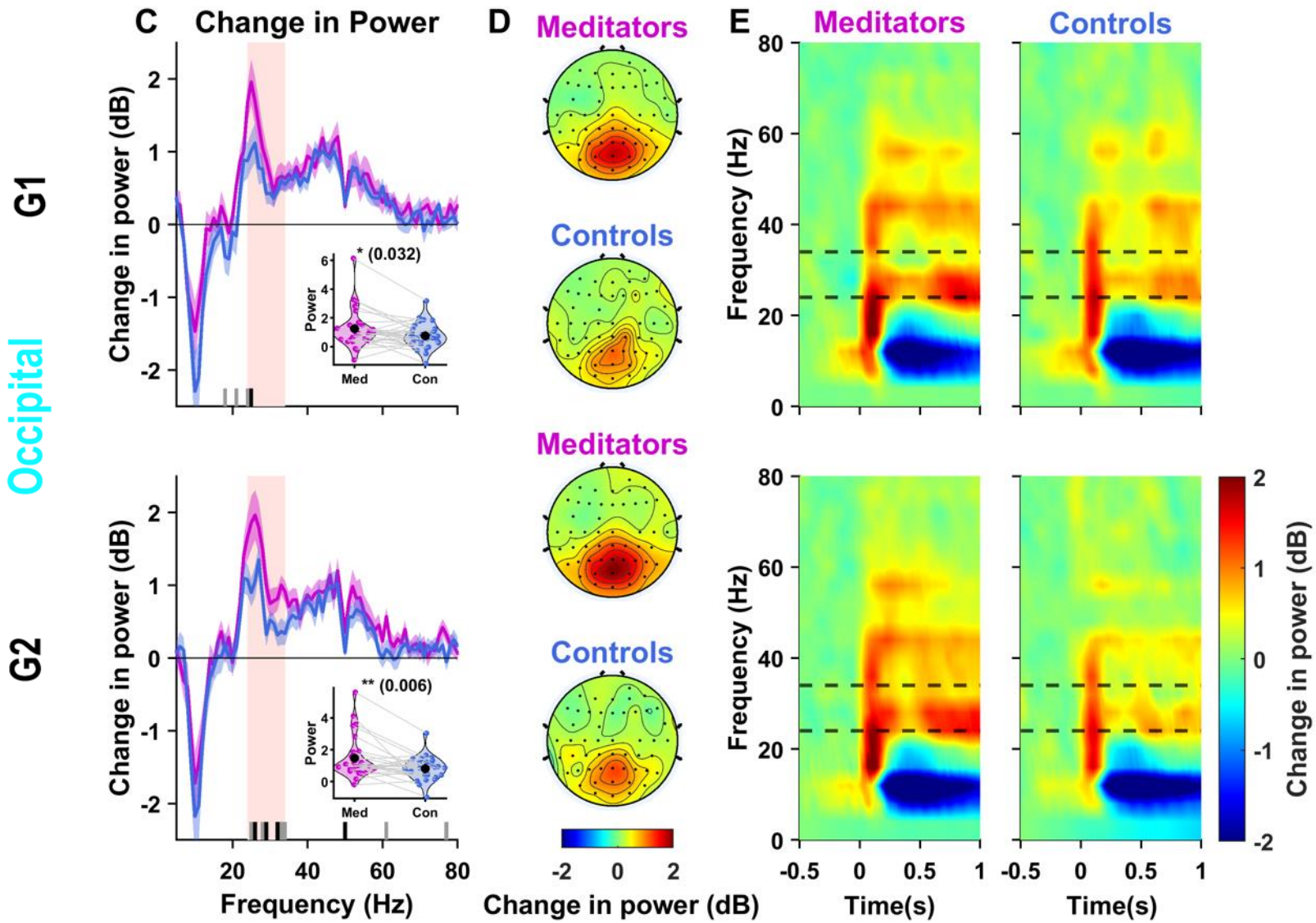

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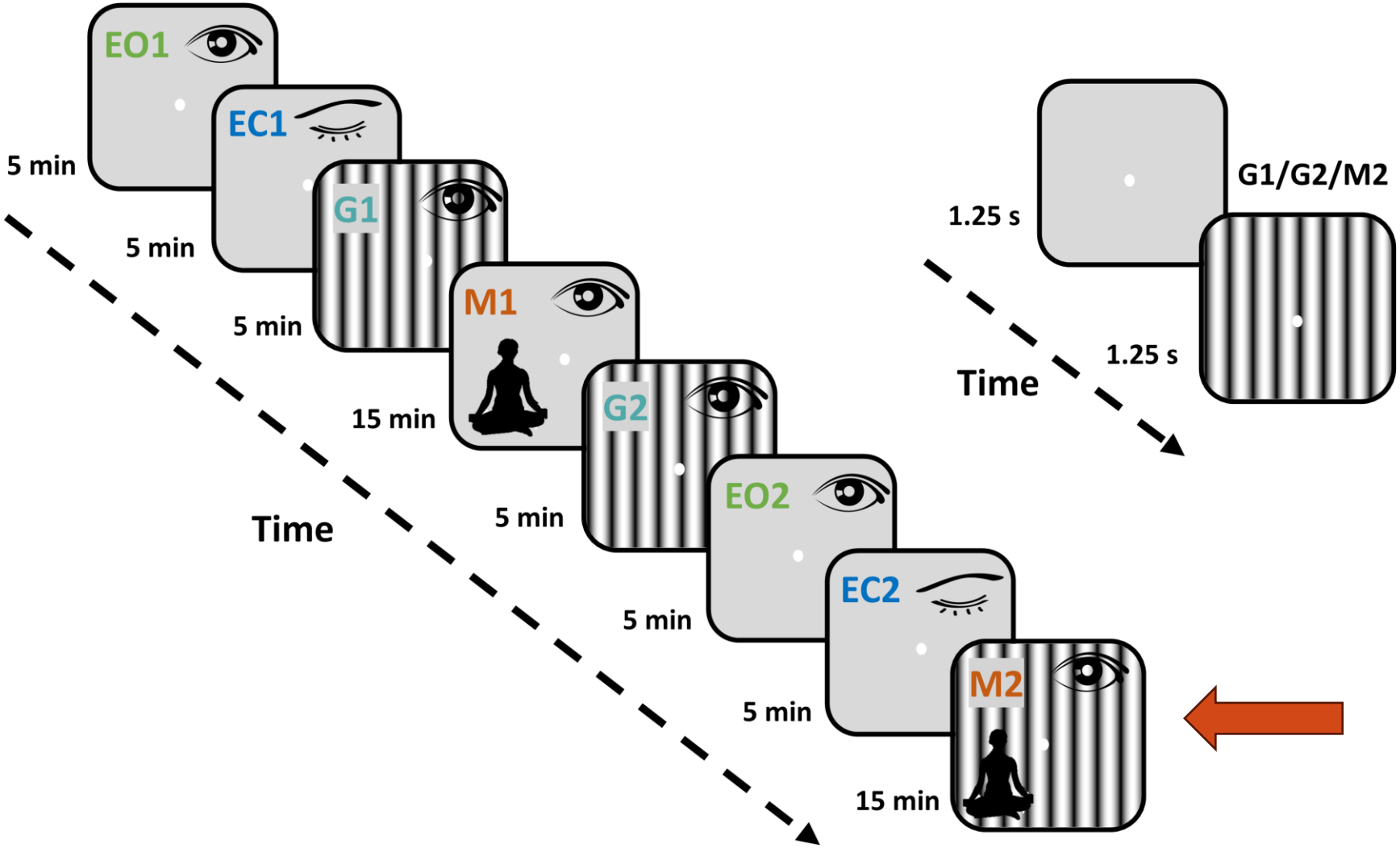
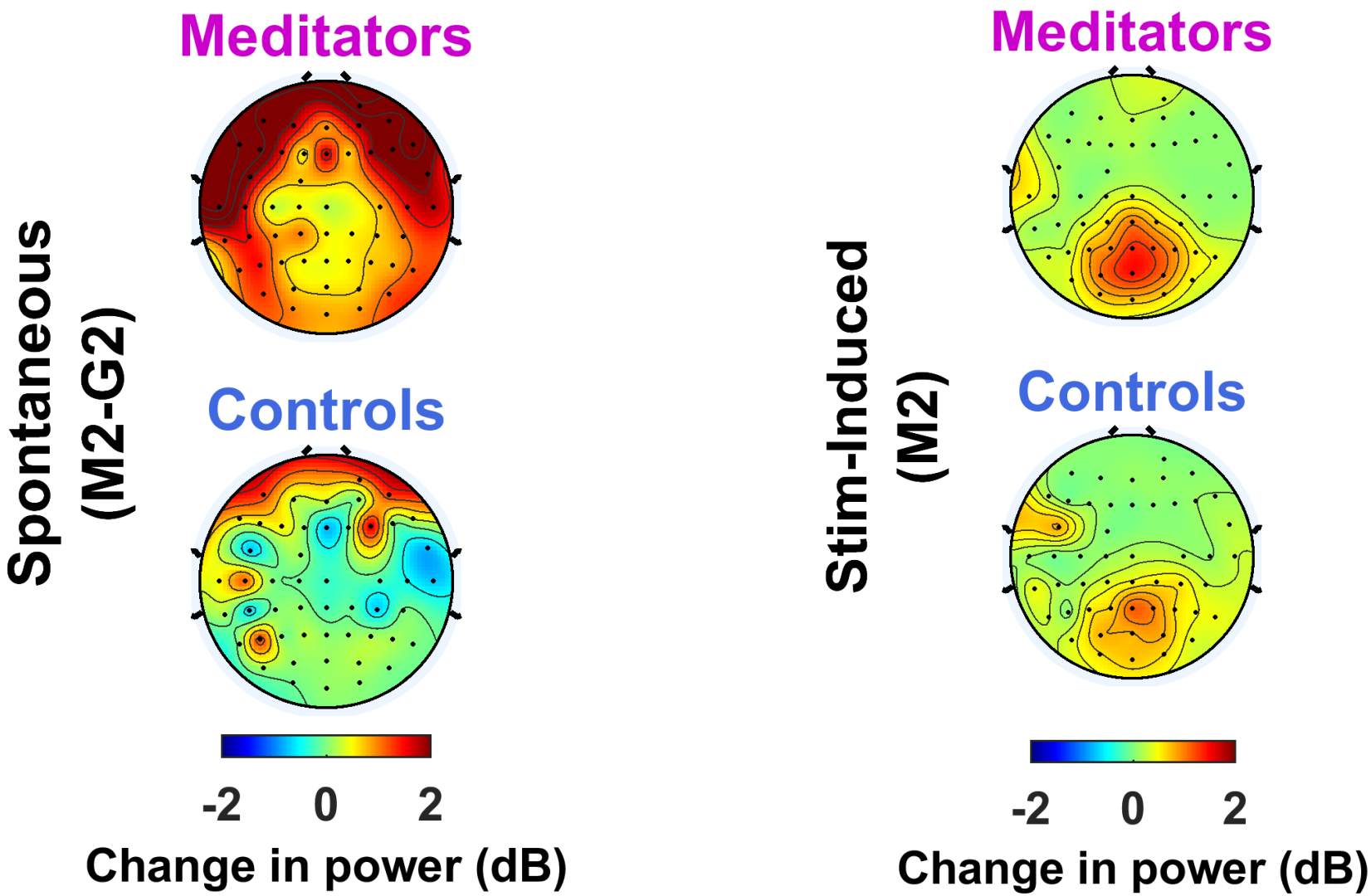
Gamma + Meditation protocol (M2)



Stimulus-free and Stimulus-induced gamma co-exist



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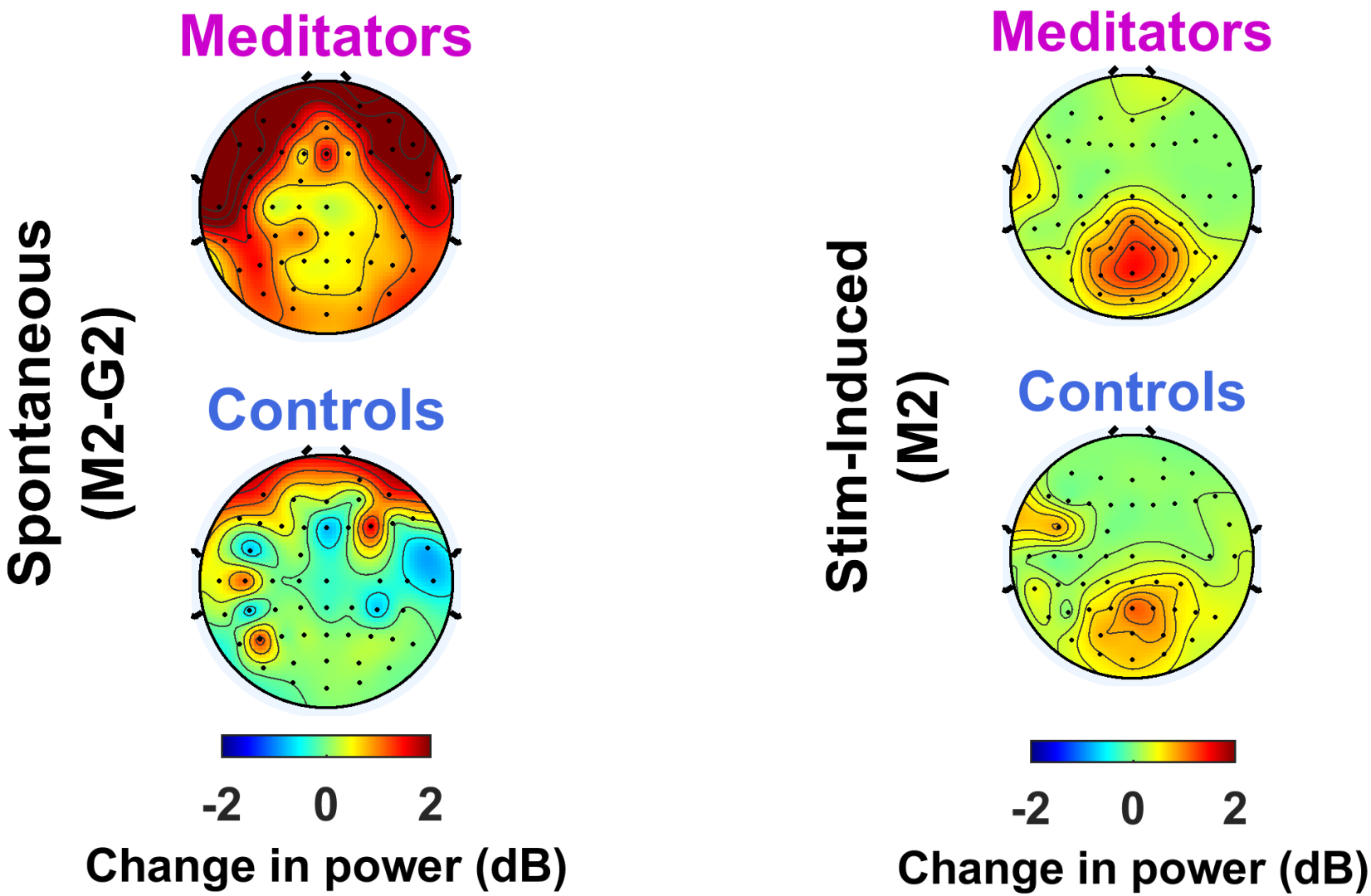
Meditation could boost the long-range inhibitory network and provide protection against ageing and neurodegeneration.



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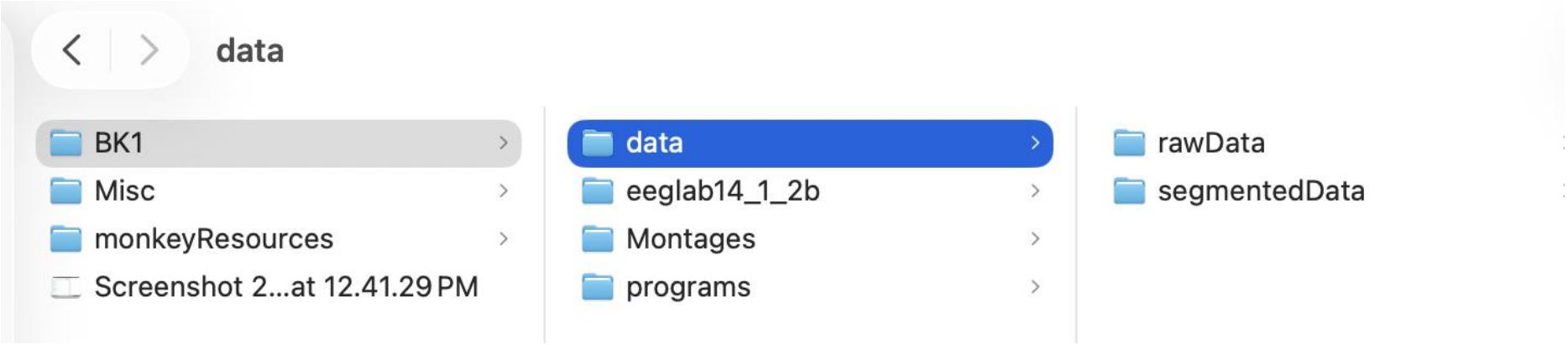
Organization of data and programs

- Create a folder called BK1. Keep the segmentedData that has been shared with you under data/segmentedData
- Programs – to be cloned from github
  - https://github.com/supratimray/ProjectDhyaanBK1Programs
  - https://github.com/supratimray/CommonPrograms
  - https://github.com/supratimray/Montages
- Software – to be downloaded
  - Chronux: https://chronux.org/
  - EEGLab: https://sccn.ucsd.edu/eeglab/
  - FieldTrip: https://www.fieldtriptoolbox.org/